**REVISION TEST****3**

Name of the Student: _____

Time: 3 hours

Class : 10th

Sub: MATHS

Sec: Lyra / Pola / Nova

Marks: 100

I. Answer all the problems.**12 x 1 = 12m**

1. State fundamental theorem of arithmetic.
2. Express $\tan \theta$ in terms of $\cos \theta$.
3. Write a polynomial having degree 2026.
4. What is the discriminant of the quadratic equation $px^2 + qx + y = 0$.
5. The first negative term of the **A.P.**

13, 9, 5, 1,is _____**[]**

a. -1

b. B. -2

c. C. -3

d. D. -4

6. Create a linear equation in two variables which is parallel to the given line

$$3x + 2y + 10 = 0.$$

7. If $\sin \theta = \frac{3}{5}$ then $\tan \theta =$ _____

8. State basic proportionality theorem.

9. A letter is chosen at random from the letters of the word "**MATHEMATICS**" what is the probability that the letter chosen is a vowel?

10. Draw a rough sketch as a pair of tangents of a circle passes through the end points of its diameter.

11. A circle can have _____ parallel tangents at the most

a. 1

b. 2

c. 0

d. infinite

12. Volume of a cube is 125 cm^3 then its side is _____.

II. Answer all the questions.**8 x 2 = 16m**

13. Establish the relationship between the zeroes and co efficient of the polynomial

$$ax^2 + bx + c. \text{ Hence find the sum and product of the zeroes of } 2x^2 - 7x + 8.$$

14. Mention the conditions for the quadratic equation $ax^2 + bx + c = 0$ to have i) equal real roots ii) no real roots.

15. Write the number of tangents drawn to a circle from a point **A** in the following cases.

- i. If A lying inside the circle.
- ii. If A lying on the circle.
- iii. If A is lying outside the circle.

16. Evaluate $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$.

17. Find the ratio in which the line segment joining the points **(-3, 10)** and **(6, -8)** is divided by **(-1, 6)**

18. Define the following terms.

- a. Angle of elevation
- b. Angle of depression

19. **2 Cubes** each of volume **64cm^3** are joined end to end find the surface area of the resulting cuboid.

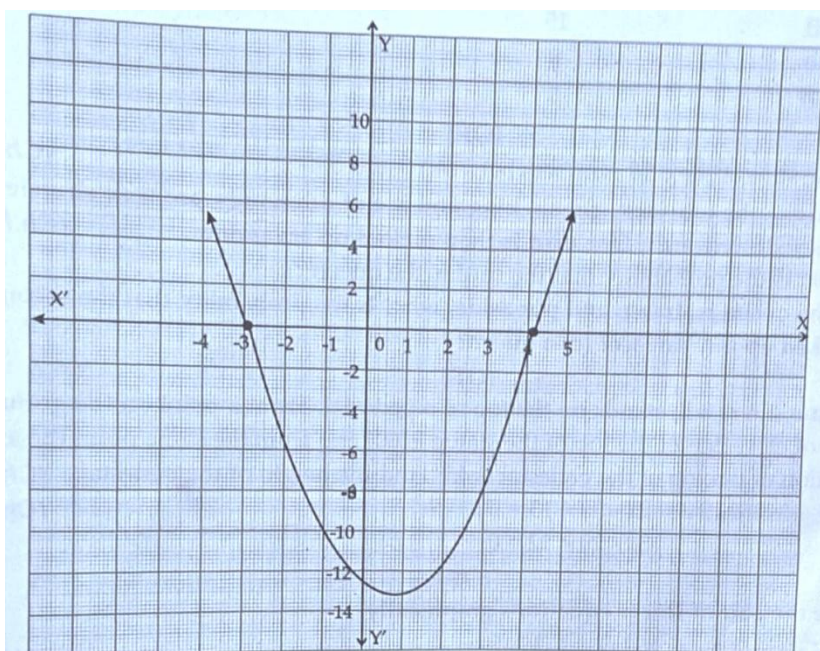
20. Give two different examples of pair of.

- a. Similar figures
- b. Non – similar figures

III. **Answer all the questions.**

$$8 \times 4 = 32\text{m}$$

21. By observing the graph answer the following questions.



- a. What is the shape of graph in the figure?
- b. How many zeroes are there for that polynomial?
- c. Find the product of zeroes of the polynomial.
- d. Write the zeros of the polynomial.

22. Reproduce $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$ as $7 + \tan^2 A + \cot^2 A$.

23. Prove that the length of the tangents drawn from an external point to a circle are equal.

24. A toy is in the form of a cone of radius **3.5cm** mounted on a hemisphere of same radius. The total height of the toy is **15.5 cm**. Find the total surface area of the toy.

25. Write the formula for mode for grouped data? Explain each term in it?

26. Two dice, one blue and one grey are thrown at the same time. Write down all the possible outcomes what is the probability that the sum of the two numbers appearing on the top of the dice is.

1. 8

2. 13

3. Less than or equal to 12.

27. Find two consecutive positive integers, sum of whose squares is **365**.

28. Check whether 301 is a term of the list of numbers **5, 11, 17, 23,**

IV. Answer all the questions. There is an internal choice for each question.

5 x 8 = 40m

29. A. is $\sqrt{5}$ irrational? Justify your answer.

(Or)

B. A girl of height 90cm is walking away from the base of a lamp-post at a speed of 1.2m/s. if the lamp is 3.6 above the ground. Find the length of her shadow after 4 seconds.

30. A. Find the co-ordinates of the points of trisection of the line segment joining (4, -1) and (-2, -3).

(Or)

B. A chord of a circle of radius 15cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle.

Use $\pi = 3.14$ and $\sqrt{3} = 1.73$.

31.A. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting.

- i. A spade
- ii. A face card
- iii. A king of red colour
- iv. A queen of diamonds

(Or)

B. a straight highway leads to the foot of a tower. A man standing at the top of the tower observe a car at an angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the time taken by the car to reach the foot of the tower from this point.

32. A. The following table gives the literacy rate (in percentage) of 35 cities. Find the mean literacy rate.

Literacy rate in %	45-55	55-65	65-75	75-85	85-95
Number of cities	3	10	11	8	3

(Or)

B. A sum of Rs. 100 is invested at 8% S.I per year calculate the interest at the end of each year. Do these interests form an A.P? If so, find the interest at the end of 30 years making use of this fact.

33.A. Check whether the pair of equation $3x + 4y = 10$, $x - y = 1$ is consistence or not solve them graphically?

(Or)

B. Solve the pair of linear equations graphically $2x - y = 4$, $x - 3y = -3$.